

MASTER'S THESIS IN PHYSICS

Integration of an SiPM-Based Scintillation Detector into the Phase-2 CMS Drift Tube Readout Chain Using the lpGBT ASIC

written by

Nils Jonathan Esper

presented to

The Faculty of Mathematics, Computer Science and Natural
Sciences of RWTH Aachen University

Institute of Physics III A

supervised by

Univ.-Prof. Dr. rer. nat. Alexander Schmidt

and as second reviewer

Univ.-Prof. Dr. rer. nat. Oliver Pooth

Aachen, December 2025

Abstract

Abstract.

Contents

1	Introduction	1
2	Theoretical background	2
2.1	The Standard Model of particle physics	2
2.1.1	Fermions	2
2.1.2	Bosons	3
2.1.3	Incompleteness and challenges	3
2.2	Properties of the muon	3
2.3	Atmospheric muons	4
2.3.1	Cosmic rays and muons	4
2.3.2	Energy spectrum and angular distribution	5
3	Principles of particle detection	7
3.1	Interaction of heavy charged particles with matter	7
3.1.1	Ionization and excitation	7
3.1.2	Bremsstrahlung and other radiative losses	8
3.1.3	Coulomb scattering	9
3.1.4	Other effects	9
3.2	Scintillators and photon detection	10
3.2.1	Scintillators	10
3.2.2	Silicon photomultipliers	11
3.3	Gaseous detectors	12
3.3.1	Operation modes of gaseous detectors	12
3.3.2	Drift chamber	13
3.4	Modern readout electronics	14
3.4.1	General aspects of detector readout	14
3.4.2	FPGA devices	14
3.4.3	Basics of logic circuit development	14
4	CMS Drift Tube system	15
4.1	Compact Muon Solenoid experiment	15
4.1.1	Large Hadron Collider	15
4.1.2	Experiment overview	15
4.2	Drift Tube system	15
4.2.1	Subdetector setup	15
4.2.2	Detector impact and performance	15
4.3	Phase-2 DT readout electronics	15
4.3.1	Motivation for upgrade	15
4.3.2	Minicrate electronics	15
4.3.3	HTG-940 backend	16
4.4	DT test stand in Aachen	16

4.4.1	MB1 DT chamber	16
4.4.2	Installation of Phase-2 minirate	16
5	Triggerless Drift Tube data analysis and track reconstruction	17
5.1	Intra-channel timing calibration with testpulses	17
5.2	Drift cell hits	17
5.3	Superlayer hit clustering	17
5.4	Superlayer cluster fitting	17
5.5	Superlayer relative time alignment	18
5.6	Grouping of superlayer cluster fits	18
5.7	Global track building	18
5.8	Reconstructed muon tracks	18
5.9	Acceptance correction and angular distributions	18
6	SiPM-based scintillation detector	19
6.1	Detector setup	19
6.1.1	Mechanical structure	19
6.1.2	SiPM electronics	19
6.2	FPGA readout system	19
6.2.1	Input circuitry	19
6.2.2	FPGA firmware overview	19
6.2.3	Signal sampling with FPGA-internal comparators	19
6.2.3.1	Working principle	20
6.2.3.2	Voltage bias calibration	20
6.2.3.3	Input timing calibration	20
6.2.4	Online signal processing	20
6.2.4.1	Timestamp assignment	20
6.2.4.2	Data funneling	20
6.3	lpGBT Mezzanine PCB	20
6.3.1	lpGBT ASIC	21
6.3.2	Mezzanine board	21
6.3.3	Backend firmware modifications	21
7	Scintillator data analysis	22
7.1	SiPM noise scan	22
7.2	SiPM hits	22
7.3	SiPM coincidences	22
7.4	Scintillator strip coincidences	22
8	Combined measurement with both detectors	23
8.1	Combined detector setup	23
8.2	Temporal hit correlation	23
8.3	Spatial muon correlation	23
9	Summary and Outlook	24
	Appendix A Additional details	25
	Bibliography	26
	List of Acronyms	30

1 Introduction

- DT cosmic test stand in Aachen
- Scintillation detector self-developed at institute
- Read out both detectors together
- Create interface to DT chamber to later be used for trigger scintillators (for studies with DT chamber) or other detectors (using DT chamber triggerless reconstructed muons as reference)
- Requirement: Synchronous operation in ns range
- Use lpGBT chip to develop own readout system to interface Phase-2 DT readout
- lpGBT: Commonly used in HL-LHC upgrade project, ensures synchronous operation

2 Theoretical background

2.1 The Standard Model of particle physics

The Standard Model of particle physics (SM) is the state-of-the-art theory model used in particle physics, describing elementary particles and their interactions [1–3]. It successfully predicts the results of many particle physics experiments to a very precise level at the reachable energy scales [4]. The following sections should provide a brief overview of the particles and fundamental forces in the SM. The predicted particles in the SM are shown in fig. 2.1. Despite its success, the current model is considered incomplete [3], which will be discussed briefly as well.

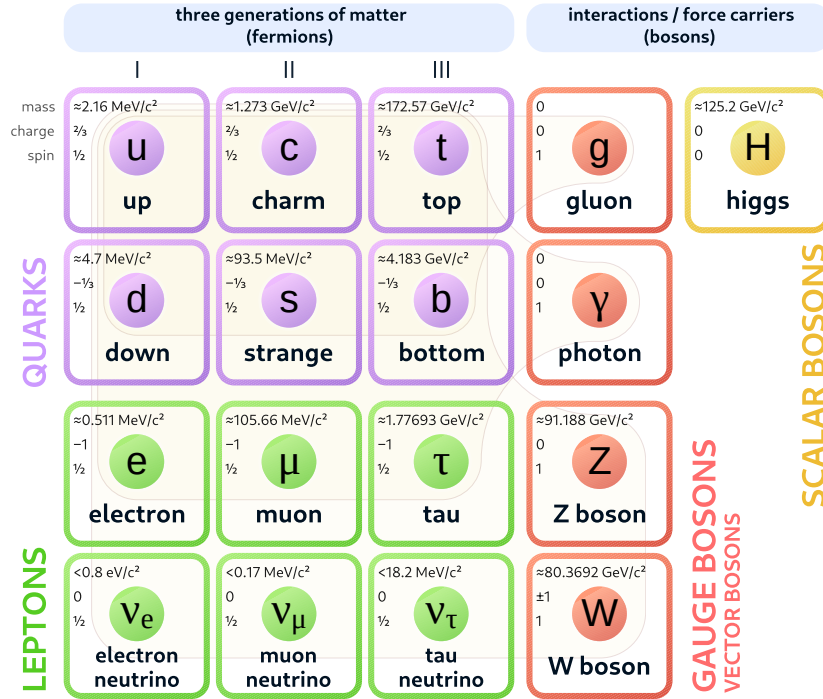


Fig. 2.1: Illustration of the particle content of the SM [5, adapted].

2.1.1 Fermions

All matter particles in the SM are fermions which have half-integer spin and obey Fermi-Dirac statistics [6, 7]. The species of a matter particle is referred to as flavor. The SM predicts three generations of spin- $\frac{1}{2}$ fermions, with four particles in each generation. Per generation there are two quarks (one up-type quark with electric charge $\frac{2}{3}e$ and one down-type quark with electric charge $-\frac{1}{3}e$) and two leptons (one charged lepton with electric charge $-e$ and one electrically neutral and massless neutrino)¹. While the particle masses increase with increasing fermion generation, other properties of the quarks and leptons from different generations are similar.

¹ $e = 1.602 \times 10^{-19} \text{ C}$ is the elementary charge, i.e. the absolute value of the electron charge [8].

The first generation fermions form stable matter, i.e. atoms consisting of protons, neutrons² and electrons, while higher generation fermions decay subsequently into lighter particles of the first generation. For all matter particles, there exists a corresponding antiparticle with similar properties except inverted charge (inverted sign of quantum numbers). Antiparticles are denoted with a bar over the particle symbol or their electric charge (with the index $^\pm$) [1, 3].

2.1.2 Bosons

There are three fundamental interactions in the SM: Electromagnetic interaction, weak interaction and strong interaction. These fundamental interactions are mediated by integer spin gauge bosons which obey Bose-Einstein statistics [9]. In fact, all four mediator particles in the SM have spin-1. The electromagnetic interaction is mediated by the photon (γ) and affects all particles which carry electric charge. The photon itself is massless and electrically neutral, giving the electromagnetic interaction an infinite range. The weak interaction is mediated by the massive $W^\pm$ and Z^0 bosons and has a short range due to the large masses of the mediator particles. It is the only interaction to allow flavor changes of particles. Electromagnetic and weak interaction were unified into the electroweak theory. The strong interaction is mediated by the massless gluon (g). All particles carrying color charge engage in strong interactions, which are the quarks and the gluon itself. While this interaction is the strongest one, it is also restricted in range due to an effect referred to as color confinement. This effect arises from the self-coupling of the gluon and implies that one expects only color neutral bound states and no free quarks in nature [1, 3]. In the electroweak theory, the force mediator bosons are initially massless. However the weak mediators are massive which was both experimentally measured but also required in order for weak interactions to be limited in range. The proposed Higgs mechanism [10, 11] resolves this issue employing spontaneous symmetry breaking. The mechanism leads to massive $W^\pm$ and Z^0 bosons and a massless photon while predicting the existence of an additional spin-0 boson, the Higgs (H) boson [3]. This particle was ultimately discovered in 2012 at CERN [12, 13].

2.1.3 Incompleteness and challenges

While all particles predicted by the SM have now been observed [1] and the model succeeds to predict their interaction properties to very high precision [4], the model is considered deficient. Firstly, the fourth fundamental interaction observed in nature, gravitation, is not included in the model. While gravitativ interaction is significantly weaker than the three interactions of the SM and therefore not relevant for high-energy physics experiments, the model remains incomplete [1]. Secondly, the observation of neutrino oscillations [3, 14] implies non-zero neutrino masses³, however in the SM the neutrinos are considered massless. Cosmic observations hint the existence of dark matter and dark energy, making up $\approx 95\%$ of the matter-energy density of the universe [3, 15]. However the SM only attempts to model the $\approx 5\%$ of baryonic matter. There are additional shortcomings of the SM as the unsatisfactory large number of parameters (26 parameters chosen from experimental observations) and the unresolved matter-antimatter asymmetry observed in nature [3]. This incomplete list should point out the motivation future developments in particle physics theory and experiments.

2.2 Properties of the muon

This thesis focuses on the development of a readout system for a muon detector. Therefore the properties of the particle of interest should be presented in this section in short.

²Protons and neutrons are composite particles of first generation quarks. Note that the free neutron is not a stable particle, however when bound in a nucleus it does not decay.

³Strictly speaking, not all neutrinos are required to be massive, since the oscillations only imply non-zero mass differences between the neutrino flavors.

Basic properties The muon (μ^-) is the second generation charged fermion. It has an electric charge of $-1e$ and a mass of $m_\mu = 105.658 \text{ MeV}/c^2$. It is therefore $\approx 207\times$ heavier than its first generation relative, the electron (e^-). Since the muon is lighter than the lightest existing hadron (the π^0 meson), it can only decay leptonically [8]⁴:

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu, \quad \mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu.$$

The muon lifetime is around $\tau_\mu = 2.196 \mu\text{s}$ [8]. Due to time dilation, high-energy muons can travel significant distances (e.g. in Earth's atmosphere, as discussed in sec. 2.3).

History of discovery The muon was initially discovered in cloud chamber experiments with cosmic rays in 1937, where a charged particle more massive than the electron other than the proton was postulated [16, 17]. However, it was initially labeled " μ -meson" and confused with the charged pions ($\pi^\pm$ mesons). These mesons have masses in the same order of magnitude as the muon and were thought to be the mediator particle in Yukawa's theory of strong interactions [18]. The leptonic nature of the muon and the distinction from the pions was only revealed later, when a decay of pions into muons could be observed [19].

2.3 Atmospheric muons

Since in this work the muon detector and its developed readout is only operated with atmospheric muons rather than with some other particle source, the creation and properties of atmospheric muons should be briefly described in this section.

2.3.1 Cosmic rays and muons

Cosmic rays are particles reaching the Earth from space. The first indication of cosmic radiation was through balloon flight experiments of Hess in 1912, who discovered an increasing intensity of ionizing radiation with altitude [20, 21]. Cosmic rays mainly consist of protons, alpha particles and heavier nuclei of a wide range of energies [22–24]. These particles can have many different origins, ranging from our solar system to our galaxy to extragalactic sources. The understanding of cosmic ray sources is still an active field of research today [23, 25].

Cosmic ray spectrum The flux of cosmic rays is highest for particle energies $\mathcal{O}(1 \text{ GeV})$ but the tail towards high energies extends beyond $\mathcal{O}(1 \text{ ZeV})$ [8]. The energy spectrum of cosmic rays is observed to approximately follow a power law with an energy-dependent spectral index γ [8]:

$$I(E) \propto E^{-\gamma}, \quad \text{where } \gamma \approx \begin{cases} 2.7, & \mathcal{O}(1 \text{ GeV}) \leq E \leq \mathcal{O}(1 \text{ PeV}) \\ 3.0, & \mathcal{O}(1 \text{ PeV}) \leq E \leq \mathcal{O}(100 \text{ PeV}) \\ 3.3, & \mathcal{O}(100 \text{ PeV}) \leq E \leq \mathcal{O}(1 \text{ EeV}) \\ 2.5, & \mathcal{O}(1 \text{ EeV}) \leq E \end{cases}.$$

Interactions with Earth's atmosphere Interactions of the primary cosmic rays with Earth's atmosphere lead to energy loss of the cosmic particles and to a change in particle composition. In strong interactions, secondary particles can be produced. The most common secondary particles are pions ($\pi^\pm$, π^0), however there are also many other particles produced, such as kaons ($K^\pm$, K^0), hyperons and nucleons [24]. Notably, cosmic rays can create large air showers in the atmosphere through cascades of interactions. This was first observed by Auger in 1939 through coincidences of particle detectors on the surface at far distances [26].

⁴Decays with an associated photon or an additional electron-positron pair (e^+e^-) are possible but strongly suppressed with branching fractions $\leq \mathcal{O}(10^{-5})$. Therefore shown decay mode is by far the dominating one [8].

Atmospheric muons Although their fraction in primary cosmic particles from space is negligible, muons have the largest fraction of charged particles arriving on Earth’s surface [22]. These muons originate from the decay of the secondary particles created in Earth’s atmosphere. Charged pions and kaons have lifetimes of $\mathcal{O}(10^{-8} \text{ s})$ [8] which is too short to reach Earth’s surface, even when accounting for time dilation. The charged pions decay to muons with a branching fraction of ≈ 0.999 [8]:⁵

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu, \quad \pi^+ \rightarrow \mu^+ + \nu_\mu.$$

Similarly the charged kaons can decay to muons with a branching fraction of ≈ 0.635 [8]:⁶

$$K^- \rightarrow \mu^- + \bar{\nu}_\mu, \quad K^+ \rightarrow \mu^+ + \nu_\mu.$$

Most muons are produced at altitudes of $\approx 10 - 15 \text{ km}$ and lose $\approx 2 \text{ GeV}$ before reaching the surface [22, 27].

2.3.2 Energy spectrum and angular distribution

In the following, the energy spectrum and angular distribution of cosmic muons on Earth’s surface should be briefly presented, as they are the baseline to calculate the expected detector response for the measurements presented in this thesis. In principle, calculations of these quantities is quite complex, due to the variety of decay processes and energy loss mechanisms that can occur. However there are useful parametrizations of the measured muon spectra which allow for a simplified model to be used for simulations or approximate predictions [27].

Energy spectrum A good approximation of the flux of the vertical (zenith angle $\theta = 0$) atmospheric muons is provided by the following formula [27]:

$$I(E, \theta = 0) \propto (E_0 + E)^{-n} \left(1 + \frac{E}{\epsilon}\right)^{-1}.$$

In this formula, E_0 is the energy loss of the muon from the point of its creation to the surface, and assumed constant. This is of course an approximation, especially for lower muon energies, as the energy loss is not really independent of the particle energy (see sec. 3.1). Before the meson decays into the muon, it also loses energy on its way towards the surface. When the energy of this particle (and consequently of the resulting muon) is higher, the meson can lose more energy before it decays. This effect is accounted for with the factor $\epsilon > 1$ [27].

Azimuth angle dependence The dependence of the muon flux on the azimuth angle ϕ is small and will therefore be neglected here. However there is a dependence on the zenith angle θ which can be parametrized as follows [27]:

$$I(E, \theta) \propto (E_0 + E)^{-n} \left(1 + \frac{E}{\epsilon}\right)^{-1} D(\theta)^{-(n-1)},$$

$$\text{where } D(\theta) = \sqrt{\left(\frac{R^2}{d^2} \cos^2 \theta + \frac{2R}{d} + 1\right)} - \frac{R}{d} \cdot \cos \theta$$

⁵There are also suppressed decay modes with associated photons [8].

⁶There are also semileptonic decay modes with associated pions with a branching fraction of ≈ 0.033 . As before, decays with associated photons are also possible but strongly suppressed [8].

This formula takes into account the variation in muon path length to the surface for different zenith angles assuming a spherical Earth. $R = 6378$ km is the radius at the Earth's surface [8]⁷ and $d = \mathcal{O}(10$ km) is the vertical path length that the muon travels [22, 27]. When approximating $R \gg d$ (which essentially translates to assuming a flat Earth), one obtains [27]

$$D(\theta) \approx 1/\cos \theta.$$

This yields the more commonly known approximation of the atmospheric muon angular dependence with a cosine power law with typical values of $n \approx 3$ (also see sec. 2.3.1) [27]:

$$I(E, \theta) \propto (E_0 + E)^{-n} \left(1 + \frac{E}{\epsilon}\right)^{-1} \cos^{(n-1)} \theta. \quad (2.1)$$

Parameter values for Aachen Using available values for the parameters of the presented approximation, the expected muon rate which is measured in this thesis should now be estimated. The available parameters in [27] are obtained from data measured in Hamburg, Germany (53.56° N, 10° E, altitude: m) which is considered reasonably close to Aachen (53.56° N, 10° E, altitude: m) to use similar values. The obtained energy and zenith angle dependency are plotted in fig. 2.2.

$\int_{1 \text{ GeV}}^{\infty} dE I(E, \theta = 0)$	n	E_0	ϵ
$(65.2 \pm 1.5) \text{ m}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$	3.00 ± 0.02	$(23.78 \pm 0.30) \text{ GeV}$	2000 (fixed)

Tab. 2.1: Obtained values for a fit the approximation in eq. 2.1 to atmospheric muon data taken at Hamburg, Germany. Taken from [27].

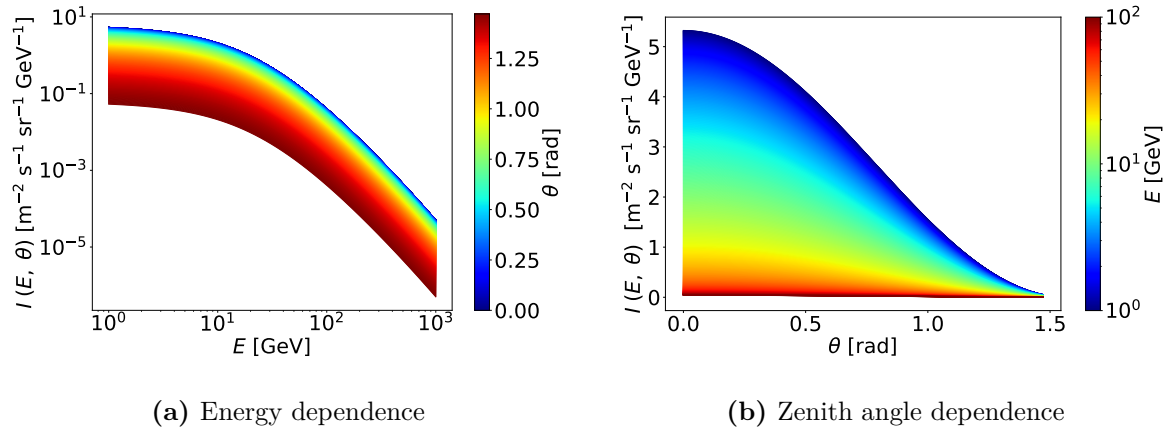


Fig. 2.2: Distributions for the atmospheric muon energy (E) and zenith angle (θ) dependence according to eq. 2.1 for the parameters in tab. 2.1.

Prediction of detector rate The expected muon rate for a detector with area A can now be obtained by integration over energy, solid angle and area

$$\dot{N} = \int_{E_{\text{thres}}}^{\infty} dE \int d\Omega \int dA I(E, \theta).$$

For this calculation, the energy detection threshold E_{thres} should be taken into account. Note that the solution of the solid angle and area integrals is not trivial for arbitrary detector geometries, since the effective detector area might be a function of the solid angle.

[Optimize this section](#)

⁷The given value is the nominal Earth equatorial radius and should serve as an approximation for the spherical Earth assumption of the formula.

3 Principles of particle detection

3.1 Interaction of heavy charged particles with matter

Understanding particle interactions with matter is essential when constructing and operating particle detectors. This is because detectors are never sensitive to elementary particles themselves, but to the signatures that the particles leave in the detector through interactions with the detector material [28].

The topic of particle interactions is quite rich, as different particles can undergo different interactions and the interaction types depend on the particle energy. Since in this thesis a muon detector is discussed, the topic is restricted to interactions of heavy charged particles (which include the muon). Particles passing through matter interact with it in various ways and thereby lose energy. A quantity commonly used in this context is the mean particle energy loss per path length in the material $-\langle dE/dx \rangle$ which also called stopping power [28]¹.

3.1.1 Ionization and excitation

Charged particles can ionize and excite atoms of the medium thereby losing energy. This effect is the dominant process of energy loss for charged particles, until radiative effects become relevant at high energies (sec. 3.1.2).

Stopping power calculation Generally, the stopping power can be described with an integral over all possible energy transfers T for the collisions between the charged particle and the shell electrons [28]:

$$-\left\langle \frac{dE}{dx} \right\rangle = n_e \int_{T_{\min}}^{T_{\max}} T \frac{d\sigma_a}{dT}(M, \beta, T) dT. \quad (3.1)$$

Here, n_e is the target electron density and $T_{\min}$ ($T_{\max}$) the minimum (maximum) allowed energy transfer. The differential cross section $d\sigma_a/dT(M, \beta, T)$ is a measure of the probability of the collision process of given energy transfer T for a particle with mass M and velocity $\beta = v/c$ ($c = 2.998 \times 10^8$ m/s is the speed of light [8]). Respecting the fact that the minimum energy transfer can not be arbitrarily small in due to the quantized electron energy levels in the atom and allowing approximations for heavy particles $M \gg m_e$ (which is the case for muons, see sec. 2.2), one obtains [8, 28]

$$T_{\min} = \frac{I^2}{2m_e c^2 \beta^2}, \quad T_{\max} = \frac{2m_e c^2 \beta^2 \gamma^2}{1 + 2\gamma m_e/M + m_e^2/M^2},$$

where $I \approx 1.7 \cdot Z^{0.85}$ eV is the mean excitation energy of the atom with atomic number Z , $\gamma = 1/\sqrt{1 - \beta^2}$ is the Lorentz factor of the heavy particle and $m_e = 511$ keV the electron mass [8].

¹Since the interactions also depend on the material density ρ , one usually weights the path length l with this quantity: $x = l\rho$. This gives the stopping power the unit MeV cm²g⁻¹ [28].

Bethe-Bloch formula Computing the integral from eq. 3.1 with the Mott cross section (which takes into account the spin of both the electron and the heavy charged particle) yields the Bethe-Bloch formula for the stopping power [8, 28]:

$$-\left\langle \frac{dE}{dx} \right\rangle = K \frac{Z}{A} \frac{z^2}{\beta^2} \left[\frac{1}{2} \ln \frac{2m_e c^2 \beta^2 \gamma^2 T_{\max}}{I^2} - \beta^2 - \frac{\delta(\beta\gamma)}{2} - \frac{C(\beta\gamma, I)}{Z} \right]. \quad (3.2)$$

Here, A is the atomic mass number of the medium, z the charge of the interacting heavy particle and $K = N_A e^4 / (4\pi\epsilon_0 m_e c^2)$ a factor including the Avogadro constant $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ and $\epsilon_0 = 8.854 \times 10^{-12} \text{ Fm}^{-1}$ the vacuum permittivity [8]. The formula is able to describe the stopping power for heavy charged particles in the velocity range $0.1 \lesssim \beta\gamma \lesssim 1000$ [8]. Generally, the stopping power curve behaves like $\propto 1/\beta^2$ for lower velocities and then reaches a minimum at around $\beta\gamma \approx 3.5$. Particles in this stopping power minimum are called Minimum ionizing particles (MiPs). For higher velocities, the stopping power increases logarithmically $\propto \ln \gamma$ [8, 28].

Correction terms Eq. 3.2 contains two additional correction terms (δ and C) which should be discussed in the following.

When the velocity of the heavy charged particle is as low as the shell electron velocities, the approximative description of the energy loss with the mean excitation energy is no longer valid, since quantum mechanical interference effects start to become relevant. The stopping power deviates from the $\propto 1/\beta^2$ behavior and decreases again towards lower velocities. This so-called shell correction is described by the $C(\beta\gamma, I)/Z$ term and increases the validity range of eq. 3.2 to velocity ranges below $\beta\gamma < 0.1$ [8, 28].

For high particle velocities, the stopping power also deviates from the $\propto \ln \gamma$ behavior, due to polarization effects of the material and the fact that the electric field changes its shape for relativistic particle velocities. The density correction $\delta(\beta\gamma)$ accounts for this effect, leading to a flattened rise of the stopping power curve [8, 28].

Statistics of energy loss The Bethe-Bloch formula (with its corrections) describes only the mean value of the energy loss per unit length, statistically the energy loss follows a Landau distribution. Larger energy transfers can lead to high-energy electrons, which are referred to as δ -electrons [8, 28].

Add Bethe-Bloch plot and maybe Landau distribution plot

3.1.2 Bremsstrahlung and other radiative losses

According to electrodynamics, accelerated charges emit photon radiation. In the case of charged particles in matter, the electric field is the Coulomb field of the material atom's nuclei. The radiated photons are referred to as Bremsstrahlung and lead to energy loss of the deflected particle.

Bremsstrahlung for electrons The energy loss due to Bremsstrahlung becomes dominating for particles with high velocities $\beta\gamma \gtrsim 1000$ and can be described as follows [8, 28]:

$$-\left\langle \frac{dE}{dx} \right\rangle = -\frac{E}{X_0}. \quad (3.3)$$

Here, X_0 is the radiation length, i.e. the path after which a particle has only left $1/e$ of its initial energy due to the radiative energy loss. Calculation of the radiation length is extensive and usually includes approximations. From eq. 3.3 it is clear that the particle energy decreases exponentially when radiating Bremsstrahlung photons [28]:

$$E(x) = E_0 e^{-x/X_0}.$$

Radiative losses for heavier particles Note that the description above is for electrons. Of course, heavy charged particles (such as muons) show radiative losses as well, however at even higher energies, since radiative losses are heavily dependent on the particle mass $dE/dx \propto 1/M^2$ [28]. Apart from photon radiation, other effects such as e^+e^- pair production taking place directly in the Coulomb field and photonuclear interactions have to be taken into account for such calculations [29].

Critical energy The energy at which the radiative losses exceed the ionization losses is called critical energy and depends on the particle type as well as the material. Due to the mass scaling of radiative losses, the critical energy of electrons $E_c = \mathcal{O}(10 \text{ MeV})$ is significantly smaller than the critical energy of muons $E_{\mu,c} = \mathcal{O}(100 \text{ GeV})$ [28].

Add plot for radiative losses and critical energy

3.1.3 Coulomb scattering

Charged particles can experience Rutherford scattering on the Coulomb field of the atom's nucleus. Scattering is also possible in the field of the shell electrons, however this contribution is neglectable for high particle energies $E \gg m_e c^2$. The Rutherford cross section can describe a single scattering process at angle θ for a particle with momentum p [28]:

$$\frac{d\sigma}{d\Omega} \propto z^2 Z^2 \frac{1}{p^2 \beta^2} \frac{1}{\sin^4 \frac{\theta}{2}}.$$

Total scattering angle Several scattering processes occur when the particle passes through a material of thickness x . The total scattering angle distribution resulting from the consecutive scattering processes can be approximated with a Gaussian distribution $f(\theta)$ centered around $\theta = 0$. The width θ_0 of the distribution of scattering angles can be estimated with [28]

$$\theta_0 \approx \frac{\sqrt{2\pi} m_e c}{\sqrt{\alpha}} \frac{1}{p\beta} \sqrt{\frac{x}{X_0}}$$

where X_0 is the radiation length of the material (see sec. 3.1.2) and $\alpha \approx 1/137$ is the electromagnetic fine structure constant [8].

Add plot for multiple scattering

3.1.4 Other effects

Charged particles can experience additional processes when passing through matter, which however are of less importance in the scope of this thesis. For completeness, they should still be mentioned here.

Cherenkov radiation Charged particles passing through matter with a velocity exceeding the velocity of light in the medium emit so-called Cherenkov photons. The energy loss due to this process is negligible, however some particle detectors exploit the process by detecting the emitted photons. The angle θ_c at which the photons are emitted in this process depends on the refraction index of the medium n and the particle velocity β , allowing for velocity measurements of non-relativistic particles with this process [8, 28]:

$$\cos \theta_c = \frac{1}{n\beta}.$$

Transition radiation Another effect is transition radiation, which implies the emission of photons when a charged particle crosses the boundary of different media. Also this process does not contribute significantly to energy loss. Note however that the radiated energy depends on the particle's Lorentz factor γ , allowing to exploit this effect for velocity measurements of relativistic particles [8, 28].

3.2 Scintillators and photon detection

One of the two detectors used in this work is a scintillation detector. Therefore the working principles of scintillators as well as their readout should be briefly introduced in this section.

3.2.1 Scintillators

A scintillator is a material which emits (optical) photons when a passing particle deposits energy in it. The emitted light follows a characteristic spectrum caused by the material properties, one also speaks of luminescence [28]. Scintillation detectors played an important role in the early days of particle physics, as they were used in the Rutherford scattering experiments of α -particles with nuclei. While the scintillation light was initially observed with the bare eye, the invention of the Photomultiplier tube (PMT) and in more recent days the Silicon photomultiplier (SiPM) (sec. 3.2.2) allowed scintillators to be used in modern experiments [30].

Scintillator properties Generally, a scintillator should exhibit a high efficiency for conversion of deposited energy into scintillation light, as well as a linearity between the number of scintillation photons and the deposited energy. Also the time between energy absorption and photon emission should be short in order to have a fast detector. Apart from this, the optical properties of the scintillator should be such that the scintillator is transparent for its own scintillation light and the optical coupling to the photon readout does not cause large losses [28].

Scintillator types One differentiates organic and inorganic scintillators by their material and scintillation mechanism. An inorganic scintillator consists of inorganic materials, which form a crystalline structure. The scintillation effect can be explained with the band structure of the electrons in the crystal. Energy deposition in the material can lead to excitation of an electron, which can then deexcite at a luminescence center in the crystal under photon emission. Usually these luminescence centers are impurities added to the crystal via doping [28, 30].

Organic scintillators on the other hand consist of organic materials, i.e. carbon compounds where the electronic structure of the molecules drives the fluorescent properties. Of highest relevance are the delocalized π -orbital electrons. Electrons become elevated to higher energy levels due to the energy deposition in the material and can deexcite radiatively, leading to scintillation light. A frequently used form of organic scintillators are so-called plastic scintillators, which embed the scintillation materials embedded in a solid plastic solvent [28, 30].

Photon readout To achieve high photon detection efficiency, it is important to optimize the optical connection between the scintillator and the photon detector. A special emphasis should be put on material transitions in the optical path (ideally similar refractive indices) as well as the geometry to reduce photon loss. If the wavelength of the light emitted by scintillator deviates from the ideal wavelength for the photon detector, one can also employ wavelength shifting materials which are able to convert photon wavelengths [28, 30]. There exist several devices for optical readout with (nearly) single-photon sensitivity, among them PMTs and SiPMs. The latter should be discussed in more detail in the following section, as it is used in the detector presented in this work.

3.2.2 Silicon photomultipliers

Silicon photomultipliers (SiPMs) are semiconductor devices capable of detecting single photons in the optical spectrum. Even though their name hints a similarity to PMTs which were mostly used until recent times, they do not share the main characteristics. While PMTs rely on a photocathode for photon-to-electron conversion (exploiting the photoelectric effect) and a vacuum tube housing a high-voltage electron amplification assembly to amplify the signal of single photoelectrons, the SiPM is purely a solid state device [28].

p-n-junction A diode consists of a junction of a *p*- and *n*-doped semiconductor. Doping means that defect atoms are added into the semiconductor lattice which increase the concentration of electrons (*n*-doping) or holes² (*p*-doping). At the *p-n*-junction, charge carriers drift towards the oppositely doped region because of the different densities of electrons and holes. Due to the oppositely acting electric field created by the charge displacement, a force equilibrium is reached. The result is a so-called depletion zone at the junction without any charge carriers and an intrinsic electric field. Usually one applies a reverse bias voltage to the junction, increasing the depletion zone width and electric field strength [28, 30].

Geiger-Avalanche photodiode Photons can excite electrons from the valence into the conduction band, which is referred to as internal photoeffect.³ The excitation results in an electron-hole pair in the semiconductor. In the depletion zone, the charges are separated and drift in opposite directions due to the electric field. The resulting current can be measured, making the device a photodiode capable of detecting optical photons [28].

Modifications in the semiconductor doping profile as well as higher doping concentration enable amplification of the electron and hole currents (which were created by the arriving photon) in the depletion region. When applying high enough reverse bias voltages, the photodiode becomes a Geiger-Avalanche photodiode (GAPD). Its amplification $> \mathcal{O}(10^5)$ is so strong that the depletion zone breaks down (i.e. becomes conductive) once one electron-hole-pair was created in the depletion region. This gives the GAPD the capability of single-photon detection. A resistor is needed to quench the discharge in order to "rearm" the device. The name "Geiger" is chosen in analogy to the respective operation mode of gaseous detectors which exhibit similar behavior (see sec. 3.3.1) [28].

SiPM device An SiPM consists of an array of $\mathcal{O}(10^3)$ GAPDs and their respective quenching resistors integrated onto one semiconductor die. Advantages of the diode array compared to a single larger-area diode are the reduced noise rate as well as the reduction of dead time effects, since not the full detector but only the hit array cell is disabled during the dead time. However the array structure leads to an additional effect, namely the possibility of crosstalk between cells. An additional effect are afterpulses which occurs due to charge carriers being trapped during the amplification avalanche and only being released at a later time [28, 31].

Since the diodes operate in Geiger mode, the signal amplitude is approximately similar for each single hit cell. This means that the resulting signal from the SiPM is discrete in amplitude. One refers to the signal amplitude of one cell as photo-electron equivalent (p.e.) [28, 31].

Add schematic drawing and photo of SiPM

Thermal noise As a semiconductor detector, the SiPM is subject to thermal noise. Thermally excited electrons in the sensitive region of a diode lead to triggering of one cell and thus a noise hit. In principle, the noise hits of all cells are independent. Taking into account the probability

²Positive charges in the lattice are called holes and effectively describe a missing electron in a lattice atom.

³In contrast, the external photoeffect describes the ionization of atom, i.e. freeing of an electron from an atom, due to photon absorption. This usually takes place when the photon energy is higher.

for crosstalk however, the noise spectrum of an SiPM does showcase noise hits with several p.e. amplitude.⁴ The frequency of multiple p.e. noise hits compared to single cell noise hits allows estimation of the crosstalk probability of the device [28, 31]. Note that for each additional hit cell due to crosstalk, the crosstalk probability needs to be added as a factor to the expected rate. Therefore one expects an exponentially falling behavior of the noise rate w.r.t. the noise hit amplitude.

Usually, one sets a threshold of several p.e. for hit discrimination, in order to suppress lower-order noise hits. Of course this also implies that the real event is required to have sufficient amount of photons to trigger the respective number of cells [28].

Add SiPM noise plot

3.3 Gaseous detectors

The second detector type used in this thesis is a gaseous detector, more specifically a drift chamber. Gaseous detectors rely on ionization (similar to scintillators, see sec. 3.2), however the detector material is a gas. A passing particle loses energy in the gas while ionizing gas atoms, creating electron-ion-pairs. When applying an electric field to the gas volume, the created charges are separated and a current is measured [28].

Brief history of gaseous detectors Gaseous detectors have a long history in particle physics, marking the first step from scintillation detectors read out manually "by eye" (sec. 3.2.1) towards detectors producing electronic signals [32]. The first gaseous detector was described by Rutherford and Geiger in 1908 and was able to detect α -particles from a radioactive source [33]. The Geiger-Müller tube introduced in 1928 showcased much stronger gas amplification [34]. The introduction of Multiwire proportional chamber (MWPC) by Charpak in 1968 opened new application fields of gaseous detectors [35]. The discovery of the $W^\pm$ and Z^0 bosons at the $Spp\bar{p}S$ collider at CERN in 1983 [36] was (in part) enabled by the UA-1 detector [37] consisting of a gaseous drift chamber for tracking and drift tube muon chambers.⁵ While semiconductor detectors nowadays showcase better energy and spatial resolution and have therefore replaced gaseous detectors in particle tracking detectors in many cases, gaseous detectors continue to be used in numerous applications. This is especially the case for applications which require large detector areas, where gaseous detectors captivate with reasonable prices [28]. An example for this are the muon systems of both the CMS [39] and ATLAS [40] experiments at CERN which consist solely of gaseous detectors. Also, research in the field of gaseous detectors is ongoing, which already led to novel high-performance micropattern gaseous detectors technologies [32].

3.3.1 Operation modes of gaseous detectors

Depending on the voltage applied to the detector and the resulting field strength, the gas detector operated in different modes, which should be described in this section.

Add gaseous detector operation modes plot

Recombination region When no voltage or very low voltages are applied to the gas detector, the electron-ion-pairs created by the passing ionizing particle can recombine before they can be collected by the readout electrodes [28]. This mode is therefore of no use for a particle detector.

⁴Of course it is (in principle) also possible for noise hits of two cells in the SiPM to happen independently within a short amount of time.

⁵The UA-1 muon drift chambers, which played a crucial role in the boson discoveries, were developed at Physics Institute III A in Aachen [38].

Ionization chamber For medium voltages, the primary electron-ion-pairs are collected by the electrodes. Since the energy loss of particles in gases is rather low, the number of collected charges per particle is also small.⁶ Therefore ionization chambers are rather suited for particle flux measurements than for single-particle detection [28].

Proportional region When the electric field strengths become larger with increasing voltages, the effect of gas amplification starts to play a role. The drifting electrons gain sufficient energy to ionize gas atoms themselves, leading to secondary electron-ion-pairs which themselves can again trigger ionization processes. For a wide voltage range, the resulting charge collected at the electrode remains proportional to the initially deposited charge. The gain G is defined as ratio of initial and final number of charge carriers and reaches values of $G \approx 10^3 - 10^5$ in the proportional region [28].

Region of limited proportionality When the voltage is further increased, the proportionality between initial and final number of charges becomes invalid. This is due to space charge effects, i.e. clouds of slower moving ions which shield the electric field [28].

Geiger region In the Geiger region, the output signal becomes independent of the number of primary charges. Such a detector can be used for particle counting purposes, however the ability to estimate the particle's energy loss with the deposited charge due to ionization is lost. In the Geiger mode, the gas discharge is not confined towards the electrode, but extends over the whole gas volume. Such a detector needs to be quenched (i.e. the discharge needs to be stopped) with an external resistor in order to become sensitive again to detect the next particle. The amplification in this mode can reach values of $G \approx 10^8 - 10^9$ [28].

Discharge region When the voltage is increased even beyond the Geiger region, gas discharges occur where both electrodes become connected via a plasma channel. This discharge can be triggered by ionizing particles or spontaneously. With measures to suppress the complete gas discharge, this mode can also be used for a particle detector, e.g. in so-called streamer chambers [28].

3.3.2 Drift chamber

A drift chamber is a gaseous detector which not only allows detection of ionizing particles, but enables estimations of the traversing particle's position within the gas volume. It is usually operated in proportional mode, however the amplification region where the gas amplification takes place is localized and there exists a significant volume where the created charges drift without any amplification. They can be seen as further development of the MWPC detector type. Drift chambers are famously used in muon detection systems which usually are located on the outer shell of a general-purpose particle detector experiment and therefore require large detector areas while still demanding good spatial resolution [28].

Add picture of drift chamber and drift cell

Drift region The drift region is considered the sensitive region of the detector and usually comprises most of its volume. When a passing particle creates electron-ion-pairs in this region, the charges are separated through an applied electric field. In the drift region, the field strengths are not high enough to trigger gas amplification. Instead, the charges drift with a constant drift velocity v_D , experiencing a force equilibrium between the electric force and a decelerating force due to scattering processes with other gas atoms. Note that this drift velocity depends on

⁶In argon, a MiP generates only $\mathcal{O}(100)$ electron-ion-pairs per cm, equivalent to created charges in the fC range which are (almost) impossible to detect without amplification [28].

the electric field, the gas mixture as well as the particle type.⁷ Typical drift velocities for drift chambers lie in the range $\mathcal{O}(50 \mu\text{m}/\text{ns})$ [28].

At some point the drifting charges reach the electrode and are collected. One should note that the drift with a known velocity v_D allows to estimate the drift distance of the charges and thus the initial position of the passing particle [28]:

$$x_D = \int_{t_{\text{start}}}^{t_{\text{stop}}} v_D(t) dt.$$

This is possible with only a time measurement, which is however required to have good resolution $\mathcal{O}(1 \text{ ns})$ if resolutions of $\mathcal{O}(100 \mu\text{m})$ should be reached. Note that a strong advantage of drift chambers is a spatial resolution which significantly exceeds the granularity of the readout channels. Ideally, the drift velocity is constant over the full width of the drift region, resulting in a linear space-time relation. This requires design efforts to make the electric field in the drift region as homogeneous as possible [28].

Gas amplification Drift chambers usually have an anode wire for electron collection. Usually, the field strengths close to the wire are strong enough to enable the amplification of the arriving electrons, thereby increasing the output signal amplitude. Usually the drift and amplification regions are not strictly separated, as in the described example of the anode wire. This leads to non-linearities in the space-time relation, which create the need for calibrations [28].

3.4 Modern readout electronics

3.4.1 General aspects of detector readout

3.4.2 FPGA devices

3.4.3 Basics of logic circuit development

⁷The drift velocity of electrons is significantly higher than the velocity of ions [28].

4 CMS Drift Tube system

4.1 Compact Muon Solenoid experiment

4.1.1 Large Hadron Collider

- Image of CERN accelerator complex
- Key facts: Luminosity, collision center-of-mass energy

4.1.2 Experiment overview

- CMS rendering
- Brief listing of all subdetectors

4.2 Drift Tube system

4.2.1 Subdetector setup

- DT chambers in CMS: Wheels, sectors, stations
- Components of one DT chamber: 2-3 superlayers, 4 layers, drift cells
- Gas mixture, HV values

4.2.2 Detector impact and performance

- Relevance of muon detection (e.g. dimuon Higgs decay)
- DT performance (efficiency and active channels)
- DT system resolution

4.3 Phase-2 DT readout electronics

4.3.1 Motivation for upgrade

- High-Luminosity LHC
- Increased pileup and tightened radiation environment

4.3.2 Minicrate electronics

- Minicrate location on chamber
- Minicrate functionalities
- On-Board Electronics for Drift Tubes boards

4.3.3 HTG-940 backend

- Prototype backend (AMD Xilinx Virtex UltraScale+ FPGA)
- Production backend: Serenity and BMTL1 (not available yet)
- Backend firmware (mention own modifications)

4.4 DT test stand in Aachen

4.4.1 MB1 DT chamber

- MB1 DT chamber from production in 2000s
- Reactivated infrastructure

4.4.2 Installation of Phase-2 minicrate

- Minicrate on DT chamber
- Detailed setup and channel mapping

5 Triggerless Drift Tube data analysis and track reconstruction

5.1 Intra-channel timing calibration with testpulses

- Importance of timing calibration in DT
- Refer to CMS DT calibration procedure
- Show testpulse response
- Discuss different testpulse cable length in theta SL
- Show derived timing corrections for individual channels

5.2 Drift cell hits

- Show hit occupancies of all wires
- Give list of masked (noisy) and dead wires
- Discuss rate
- Highlight and discuss feature of secondary hits in range 0-400 ns
- Apply dead time for individual channels

5.3 Superlayer hit clustering

- Triggerless reconstruction: Need to consider all recorded hits as potential muon signature
- Cluster hits within maximum drift time window
- Pattern rate discussion
- Compare to rates of fake patterns (to estimate noise influence)

5.4 Superlayer cluster fitting

- Assume constant drift velocity, straight muon track
- Perform linear fit for all laterality combinations with parameters: Arrival time, position in cell, track inclination
- Select laterality combination with best goodness of fit
- Discuss fit results
- Apply appropriate goodness of fit cut

5.5 Superlayer relative time alignment

- Mention relative alignment problem of OBDTs
- Show double peak in time difference between SL fits
- Establish relative time correction

5.6 Grouping of superlayer cluster fits

- Clustering algorithm creates multiple clusters in case of additional nearby hits
- Cope this by grouping together clusters within certain time window within same superlayer

5.7 Global track building

- Select cluster groups within time window from all superlayers
- Only use cluster groups with no ambiguity (only one cluster per SL)
- Ensure matching tracks in both phi superlayers
- Calculate 3D track from 2D projections from theta and phi superlayers

5.8 Reconstructed muon tracks

- Discuss muon rate
- Show angular distributions (with solid angle correction)

5.9 Acceptance correction and angular distributions

- Establish geometric simulation of DT chamber with simulated muon tracks
- Derive (rudimentary) geometric acceptance correction from simulation
- Apply acceptance correction to data
- Compare to expected cosmic muon angular distributions

6 SiPM-based scintillation detector

6.1 Detector setup

Mention self-development of detector at institute [41] [42].

6.1.1 Mechanical structure

- 2x 16 plastic scintillator strips with wavelength-shifting fibers
- Dual SiPM readout for each strip

6.1.2 SiPM electronics

- SiPM front-end board with preamplifier
- Backbone of detector with bias voltage supply
- Raspberry Pi for backbone control

6.2 FPGA readout system

- Mention completely new hardware and firmware development in scope of this thesis, based on previous experience and efforts of our group
- AMD Xilinx UltraScale+ ZynqMPSoC
- Control hardware and firmware from Petalinux running on ARM core of ZynqMPSoC
- Show scheme / overview of readout system

6.2.1 Input circuitry

- Synchronous testpulse source
- SiPM signal input and testpulse input (AC coupled)
- Individually selectable threshold for all channels
- (Show screenshot from SPICE simulation)

6.2.2 FPGA firmware overview

Mention fully self-developed readout firmware.

6.2.3 Signal sampling with FPGA-internal comparators

- Comparators for input signal discrimination, however no conventional, dedicated hardware but FPGA-internal comparators used.
- Use FPGA-internal blocks for deserialization to increase time resolution

6.2.3.1 Working principle

- FPGA-internal comparators (intended for differential signalling)
- Novel concept: Use FPGA-internal comparators with analog signals [43] [44] [45]
- Voltage limits of inputs
- Deserializers for higher time resolution

6.2.3.2 Voltage bias calibration

- FPGA-internal comparators (on FPGA silicon die) are subject to production uncertainties
- Unfolds in voltage bias on comparator switching point
- Scan input voltage range while keeping other comparator input constant to determine switching point of comparator (Metastability calibration)
- Account for measured biases when setting thresholds

6.2.3.3 Input timing calibration

- Bank-specific calibration of deserializers on each power-cycle leads to non-deterministic signal delays
- Use simultaneous testpulses to measure these delays
- Use delay module in firmware to correct for these effects already on the device
- Mention ringing of testpulse signal in circuit, but explain that no negative effect on calibration

6.2.4 Online signal processing

6.2.4.1 Timestamp assignment

- Hit arrival time translated to (absolute) timestamp
- Same format as ODBT hits

6.2.4.2 Data funneling

- Limited bandwidth: Can not transmit hits of all channels together
- However do not want to lose hits
- Solution: Introduce individual buffer (FiFo) for all channels
- Dynamically cycle through and activate outputs of different channel buffers
- Reduces possible hit rate compared to no funnel

6.3 lpGBT Mezzanine PCB

- Hardware interface between scintillator readout system and DT backend
- ODBTs use lpGBT as well therefore compatible format
- Synchronous clock (with backend) supplied by lpGBT chip

6.3.1 lpGBT ASIC

- Low-power GigaBit Transceiver ASIC developed centrally by CERN
- Key facts: Data transfer rate, clock recovery
- Optical fiber interface to backend
- Electrical link (eLink) interface to frontend FPGA
- Control interface via optical fiber or wired I2C

6.3.2 Mezzanine board

- Connects eLinks between lpGBT and FPGA
- (Show bit error rates of PRBS7 test)
- Show recovered 40 MHz clock
- Hosts optical transceiver to convert signals to/from optical fiber

6.3.3 Backend firmware modifications

- Modified data flow in DT backend to accomodate reception of data via slow control link from the lpGBT Mezzanine (OBDT has dedicated trigger links for data)
- (Show received counter data by backend)

7 Scintillator data analysis

7.1 SiPM noise scan

- Scan of SiPM rates for different thresholds with developed readout system
- Show resulting curve
- Discuss visible exponential (expected) behavior
- Aliasing effects (too coarse scanning steps) hide expected steps in spectrum

7.2 SiPM hits

- Show single SiPM occupancies
- Discuss observed secondary pulses: Ringing in circuit or SiPM after-pulses

7.3 SiPM coincidences

- Coincidence of two SiPMs of same scintillator strip
- Discuss strip rate
- Show time difference between both SiPMs, discuss width / time localization
- Compare to expected fake coincidence rate

7.4 Scintillator strip coincidences

- Coincidence of two strips of two detector layers (create "pixels")
- Discuss pixel rate
- Show time difference between both strips
- Compare to expected fake coincidence rate

8 Combined measurement with both detectors

8.1 Combined detector setup

- Overview schematic of combined detector and readout setup
- Highlight shared backend and synchronized clocks

8.2 Temporal hit correlation

- Group scintillator pixel hits with reconstructed DT tracks in time window
- Discuss rate of correlated events (compared to scintillator pixel hits before correlation)
- Show time difference between DT and scintillator
- Discuss distribution, highlight the synchronization

8.3 Spatial muon correlation

- Show cumulated reconstructed DT tracks for "matched" hits between scintillator and DT
- Highlight good geometric agreement
- Mention need for alignment analysis (which in principle would be possible with this data)

9 Summary and Outlook

Summary:

- Completely new SiPM readout system developed
- Incorporating modern FPGA data acquisition and FPGA-internal comparators for analog signal discrimination
- Adapted and developed new calibration procedures necessary for FPGA readout system
- Developed lpGBT Mezzanine to interface DT backend
- Developed FPGA firmware for readout and interfacing the lpGBT
- Successful modification of DT backend firmware to interface developed readout system
- Installed and commissioned Phase-2 minicrate with DT chamber at test stand in Aachen
- Framework to analyze DT and scintillator data together
- Implemented own triggerless reconstruction code which follows the CMS DT reconstruction
- Successful correlation of measured DT tracks and scintillator hits in space and time

Outlook:

- Understand open problems
- Implement coincidence IP core (currently still in development)
- Use system to perform various studies with DT chamber, scintillator or even other detectors connected to the developed readout system

Appendix

A Additional details

Bibliography

- [1] Mary K. Gaillard, Paul D. Grannis, and Frank J. Sciulli. “The standard model of particle physics”. In: *Rev. Mod. Phys.* 71 (2 Mar. 1999), S96–S111. DOI: 10.1103/RevModPhys.71.S96. URL: <https://link.aps.org/doi/10.1103/RevModPhys.71.S96>.
- [2] S. Weinberg. “The making of the Standard Model”. In: *The European Physical Journal C* 34.1 (May 2004), pp. 5–13. ISSN: 1434-6052. DOI: 10.1140/epjc/s2004-01761-1. URL: <http://dx.doi.org/10.1140/epjc/s2004-01761-1>.
- [3] Mark Thomson. *Modern particle physics*. eng. Cambridge, 2013.
- [4] CMS Collaboration. “Stairway to discovery: a report on the CMS programme of cross section measurements from millibarns to femtobarns”. In: *Phys. Rep.* 1115 (2024), pp. 3–115. DOI: 10.1016/j.physrep.2024.11.005. arXiv: 2405.18661. URL: <https://cds.cern.ch/record/2898867>.
- [5] Wikimedia Commons. *Standard model of elementary particles*. 2019. URL: https://commons.wikimedia.org/wiki/File:Standard_Model_of_Elementary_Particles.svg.
- [6] Paul Adrien Maurice Dirac. “On the theory of quantum mechanics”. In: *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character* 112.762 (1926), pp. 661–677. DOI: 10.1098/rspa.1926.0133. eprint: <https://royalsocietypublishing.org/doi/pdf/10.1098/rspa.1926.0133>. URL: <https://royalsocietypublishing.org/doi/abs/10.1098/rspa.1926.0133>.
- [7] Enrico Fermi. “Zur Quantelung des idealen einatomigen Gases”. In: *Zeitschrift für Physik* 36 (1926), pp. 902–912. URL: <https://api.semanticscholar.org/CorpusID:123334672>.
- [8] S. Navas et al. “Review of Particle Physics”. In: *Phys. Rev. D* 110 (3 Aug. 2024), p. 030001. DOI: 10.1103/PhysRevD.110.030001. URL: <https://link.aps.org/doi/10.1103/PhysRevD.110.030001>.
- [9] Bose. “Plancks Gesetz und Lichtquantenhypothese”. In: *Zeitschrift für Physik* 26 (1924), pp. 178–181. URL: <https://api.semanticscholar.org/CorpusID:186235974>.
- [10] P.W. Higgs. “Broken symmetries, massless particles and gauge fields”. In: *Physics Letters* 12.2 (1964), pp. 132–133. ISSN: 0031-9163. DOI: [https://doi.org/10.1016/0031-9163\(64\)91136-9](https://doi.org/10.1016/0031-9163(64)91136-9). URL: <https://www.sciencedirect.com/science/article/pii/0031916364911369>.
- [11] F. Englert and R. Brout. “Broken Symmetry and the Mass of Gauge Vector Mesons”. In: *Phys. Rev. Lett.* 13 (9 Aug. 1964), pp. 321–323. DOI: 10.1103/PhysRevLett.13.321. URL: <https://link.aps.org/doi/10.1103/PhysRevLett.13.321>.
- [12] CMS Collaboration. “Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC”. In: *Physics Letters B* 716.1 (2012), pp. 30–61. ISSN: 0370-2693. DOI: <https://doi.org/10.1016/j.physletb.2012.08.021>. URL: <https://www.sciencedirect.com/science/article/pii/S0370269312008581>.

- [13] ATLAS Collaboration. “Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC”. In: *Physics Letters B* 716.1 (2012), pp. 1–29. ISSN: 0370-2693. DOI: <https://doi.org/10.1016/j.physletb.2012.08.020>. URL: <https://www.sciencedirect.com/science/article/pii/S037026931200857X>.
- [14] John N. Bahcall and Raymond Davis. “Solar Neutrinos: A Scientific Puzzle”. In: *Science* 191.4224 (1976), pp. 264–267. DOI: 10.1126/science.191.4224.264. eprint: <https://www.science.org/doi/pdf/10.1126/science.191.4224.264>. URL: <https://www.science.org/doi/abs/10.1126/science.191.4224.264>.
- [15] Virginia Trimble. “Existence and nature of dark matter in the universe.” In: *Annual Rev. Astron. Astrophys.* 25 (Jan. 1987), pp. 425–472. DOI: 10.1146/annurev.aa.25.090187.002233.
- [16] Seth H. Neddermeyer and Carl D. Anderson. “Note on the Nature of Cosmic-Ray Particles”. In: *Phys. Rev.* 51 (10 May 1937), pp. 884–886. DOI: 10.1103/PhysRev.51.884. URL: <https://link.aps.org/doi/10.1103/PhysRev.51.884>.
- [17] J. C. Street and E. C. Stevenson. “New Evidence for the Existence of a Particle of Mass Intermediate Between the Proton and Electron”. In: *Phys. Rev.* 52 (9 Nov. 1937), pp. 1003–1004. DOI: 10.1103/PhysRev.52.1003. URL: <https://link.aps.org/doi/10.1103/PhysRev.52.1003>.
- [18] Hideki Yukawa. “On the Interaction of Elementary Particles. I”. In: *Proceedings of the Physico-Mathematical Society of Japan. 3rd Series* 17 (1935), pp. 48–57. DOI: 10.11429/ppmsj1919.17.0_48.
- [19] TD Lee. “A brief history of the muon”. In: *Hyperfine Interactions* 86.1 (1994), pp. 439–453.
- [20] Victor F. Hess. “Über Beobachtungen der durchdringenden Strahlung bei sieben Freiballonfahrten”. In: *Phys. Z.* 13 (1912), pp. 1084–1091.
- [21] Claus Grupen. “The History of Cosmic Ray Studies after Hess”. In: *Nuclear Physics B - Proceedings Supplements* 239-240 (2013). Proceedings of the 9th workshop on Science with the New Generation of High Energy Gamma-ray Experiments: From high energy gamma sources to cosmic rays, one century after their discovery, pp. 19–25. ISSN: 0920-5632. DOI: <https://doi.org/10.1016/j.nuclphysbps.2013.05.003>. URL: <https://www.sciencedirect.com/science/article/pii/S0920563213003708>.
- [22] Particle Data Group et al. “Review of Particle Physics”. In: *Progress of Theoretical and Experimental Physics* 2022.8 (Aug. 2022), p. 083C01. ISSN: 2050-3911. DOI: 10.1093/ptep/ptac097. eprint: <https://academic.oup.com/ptep/article-pdf/2022/8/083C01/49175539/ptac097.pdf>. URL: <https://doi.org/10.1093/ptep/ptac097>.
- [23] Thomas K. Gaisser, Ralph Engel, and Elisa Resconi. *Cosmic rays and particle physics*. eng. Second edition. Cambridge, United Kingdom: Cambridge University Press, 2016. ISBN: 9780521016469.
- [24] Peter K. F. Grieder. *Cosmic rays at earth : researcher’s reference manual and data book*. eng. Digital print. Amsterdam [u.a.: Elsevier, 2005. ISBN: 0444507108.
- [25] Veronica Bindi, Mercedes Panizza, and Martin Pohl. *Cosmic ray physics: an introduction to the cosmic laboratory*. eng. First edition. Boca Raton, FL: CRC Press, Taylor & Francis Group, 2023 - 2023. ISBN: 1-00-318138-4.
- [26] Pierre Auger et al. “Extensive Cosmic-Ray Showers”. In: *Rev. Mod. Phys.* 11 (3-4 July 1939), pp. 288–291. DOI: 10.1103/RevModPhys.11.288. URL: <https://link.aps.org/doi/10.1103/RevModPhys.11.288>.

-
- [27] Prashant Shukla and Sundaresh Sankrith. “Energy and angular distributions of atmospheric muons at the Earth”. In: *International Journal of Modern Physics A* 33.30 (2018), p. 1850175. DOI: 10.1142/S0217751X18501750. eprint: <https://doi.org/10.1142/S0217751X18501750>. URL: <https://doi.org/10.1142/S0217751X18501750>.
 - [28] Hermann Kolanoski and Norbert Wermes. *Particle detectors : fundamentals and applications*. eng. Oxford: Oxford University Press, 2020. ISBN: 9780191890710. URL: <https://doi.org/10.1093/oso/9780198858362.001.0001>.
 - [29] DONALD E. GROOM, NIKOLAI V. MOKHOV, and SERGEI I. STRIGANOV. “Muon Stopping Power and Range Tables 10 MeV - 100 TeV”. In: *Atomic Data and Nuclear Data Tables* 78.2 (2001), pp. 183–356. ISSN: 0092-640X. DOI: <https://doi.org/10.1006/adnd.2001.0861>. URL: <https://www.sciencedirect.com/science/article/pii/S0092640X01908617>.
 - [30] William R. Leo. *Techniques for nuclear and particle physics experiments : a how-to approach ; with 256 figures, 40 tables and numerous worked examples*. eng. Second revised edition. Berlin ; Heidelberg: Springer-Verlag GmbH, 1994. ISBN: 0387572805.
 - [31] Stefan Gundacker and Arjan Heering. “The silicon photomultiplier: fundamentals and applications of a modern solid-state photon detector”. In: *Physics in Medicine & Biology* 65.17 (Aug. 2020), 17TR01. DOI: 10.1088/1361-6560/ab7b2d. URL: <https://doi.org/10.1088/1361-6560/ab7b2d>.
 - [32] H J Hilke and W Riegler. “Gaseous Detectors”. In: (2020), pp. 91–136. DOI: 10.1007/978-3-030-35318-6_4. URL: <https://cds.cern.ch/record/2743166>.
 - [33] Ernest Rutherford and H Geiger. “An electrical method of counting the number of α -particles from radio-active substances”. In: *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character* 81.546 (1908), pp. 141–161. DOI: 10.1098/rspa.1908.0065. eprint: <https://royalsocietypublishing.org/doi/pdf/10.1098/rspa.1908.0065>. URL: <https://royalsocietypublishing.org/doi/abs/10.1098/rspa.1908.0065>.
 - [34] H. Geiger and W. Müller. In: *Phys. Zeits.* 29.839 (1928).
 - [35] G. Charpak et al. “The use of multiwire proportional counters to select and localize charged particles”. In: *Nuclear Instruments and Methods* 62.3 (1968), pp. 262–268. ISSN: 0029-554X. DOI: [https://doi.org/10.1016/0029-554X\(68\)90371-6](https://doi.org/10.1016/0029-554X(68)90371-6). URL: <https://www.sciencedirect.com/science/article/pii/0029554X68903716>.
 - [36] Luigi Di Lella and Carlo Rubbia. “The Discovery of the W and Z Particles”. In: *60 Years of CERN Experiments and Discoveries*, pp. 137–163. DOI: 10.1142/9789814644150_0006. eprint: https://www.worldscientific.com/doi/pdf/10.1142/9789814644150_0006. URL: https://www.worldscientific.com/doi/abs/10.1142/9789814644150_0006.
 - [37] Alan Astbury et al. *A 4π solid angle detector for the SPS used as a proton-antiproton collider at a centre of mass energy of 540 GeV*. Tech. rep. Geneva: CERN, 1978. URL: <http://cds.cern.ch/record/319371>.
 - [38] K. Eggert et al. “Large area drift tube chambers for a muon detector at the pp-collider”. In: *Nuclear Instruments and Methods* 176.1 (1980), pp. 217–222. ISSN: 0029-554X. DOI: [https://doi.org/10.1016/0029-554X\(80\)90706-5](https://doi.org/10.1016/0029-554X(80)90706-5). URL: <https://www.sciencedirect.com/science/article/pii/0029554X80907065>.
 - [39] CMS Collaboration. “The CMS experiment at the CERN LHC”. In: *JINST* 3 (2008). Also published by CERN Geneva in 2010, S08004. DOI: 10.1088/1748-0221/3/08/S08004. URL: <https://cds.cern.ch/record/1129810>.

-
- [40] ATLAS Collaboration. “The ATLAS Experiment at the CERN Large Hadron Collider”. In: *JINST* 3 (2008). Also published by CERN Geneva in 2010, S08003. DOI: 10.1088/1748-0221/3/08/S08003. URL: <https://cds.cern.ch/record/1129811>.
- [41] Erik Ehlert. “Technical realisation of an SiPM-based muon detector”. B.Sc. Thesis. Physics Institute III A, RWTH Aachen University, 2022. URL: https://web.physik.rwth-aachen.de/user/hebbeker/theses/ehlert_bachelor.pdf.
- [42] Erik Ehlert. “Construction and Operation of an SiPM based Muon Reference Detector”. M.Sc. Thesis. Physics Institute III A, RWTH Aachen University, 2024. URL: https://www.institut3a.physik.rwth-aachen.de/global/show_document.asp?id=aaaaaaaaacvdhcay.
- [43] Daoming Xi et al. “FPGA-Only MVT Digitizer for TOF PET”. In: *IEEE Transactions on Nuclear Science* 60.5 (2013), pp. 3253–3261. DOI: 10.1109/TNS.2013.2277855.
- [44] Marek Palka et al. “A novel method based solely on field programmable gate array (FPGA) units enabling measurement of time and charge of analog signals in positron emission tomography (PET)”. In: *Bio-Algorithms and Med-Systems* 10.1 (Mar. 2014), pp. 41–45. ISSN: 1895-9091. DOI: 10.1515/bams-2013-0104. URL: <http://dx.doi.org/10.1515/bams-2013-0104>.
- [45] D. Eliseev et al. “An open-sorce IP-Core for Multi-Voltage Thresholding signal acquisition with FPGAs”. In: *Journal of Instrumentation* 19.02 (Feb. 2024), p. C02035. DOI: 10.1088/1748-0221/19/02/C02035. URL: <https://doi.org/10.1088/1748-0221/19/02/C02035>.

List of Acronyms

ATLAS	A Toroidal LHC Apparatus
CERN	European Organization for Nuclear Research
CMS	Compact Muon Solenoid
GAPD	Geiger-Avalanche photodiode
MiP	Minimum ionizing particle
MWPC	Multiwire proportional chamber
p.e.	photo-electron equivalent
PMT	Photomultiplier tube
SiPM	Silicon photomultiplier
SM	Standard Model of particle physics

Acknowledgements

Acknowledgements.